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Advanced Cementing Solutions for Ultra-Deep Slimhole Wells: Overcoming High-Temperature and Cross-Flow Challenges

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Abstract

Ultradeep, slimhole, onshore wells deeper than 20,000 ft (Shook et al. 1995) with temperatures greater than 300°F present significant cement challenges, particularly with 6-in. openhole and 4.5-in. conventional liners over extended intervals (4,200-ft openhole intervals). In this scenario, the presence of crossflow during cement hydration can cause contamination and compromised zonal isolation. As a result, a sustainable cement system with high-temperature-resistant additives, elastic agents for improved mechanical performance, and expansive components to counteract shrinkage was deployed. The barrier selected is also designed to reduce operational time and resource consumption, which contributes to lower carbon emissions during well construction.

Field results proved that the optimized cement system improved zonal isolation through mitigation of contamination effects from potential crossflow. Laboratory test results indicated improved compressive strength retention under high-temperature conditions, increased flexibility and resistance to stress-induced cement failure, and effective annular sealing through expansive properties that counteract shrinkage. Furthermore, the reduction in transition time minimized fluid migration risks, which was evident in the cement bond logs (CBLs) where bonding efficiency between the cement, liner, and formation was notably improved.

The approach used to select the slurry properties necessary to improve the barrier was sufficient to allow strengthened bond performance in the pay zones and reduce the risk of fluid migration from potential crossflow. Real-time data along with a pressure match confirmed the reliability of the hydraulic model and demonstrated that this approach provided accurate plans for this operation and future operations. These outcomes highlight the value of tailored cement systems that maximize well integrity and contribute to lower carbon emissions through efficient resource selection and proper management of risks associated with remedial operations.

An innovative approach is presented, which integrates a multifunctional cement system tailored for ultradeep, high-temperature, slimhole wells. The combination of expansive (in-situ), elastic, and high-temperature-resistant additives provides a resilient solution to critical cement challenges and helps ensure

long-term well integrity. The results contribute to the advancement of well construction best practices and provide valuable insights for the optimization of cement operations in complex drilling environments.

Introduction

Cement operations in ultradeep slimhole wells present a unique set of technical and operational challenges that require specialized design and execution strategies. In the northeast (NE) field, wells are drilled to measured depths that exceed 20,000 ft, with production sections completed using 6-in. bits and narrow annular clearances. These geometries result in extended openhole intervals that are often greater than 4,200 ft and the deployment of 4.5-in. liners anchored into previously set 9 5/8-in. casings. The reduced annular space, combined with high bottomhole static temperatures (BHST) greater than 300°F, creates a difficult environment for cement placement, displacement efficiency, and long-term zonal isolation.

One of the most critical aspects of cement operations in slimhole environments is the control of the equivalent circulating density (ECD) during slurry placement. The narrow annular geometry increases frictional pressure losses, which can elevate ECD beyond the fracture gradient of the formation and create a risk of losses or compromised cement integrity. To mitigate this, the cement slurry was engineered with optimized rheological properties to allow controlled placement and minimize hydraulic risks.

The cement system used in these wells was formulated with a density of 16.0 lbm/gal and included a pre-set expansion agent to improve sealing performance and reduce transition time. Laboratory tests under simulated downhole conditions evaluated thickening time, compressive strength, and compatibility with drilling fluid. Mechanical properties, such as Young's modulus (YM) and Poisson's ratio (PR) were assessed to confirm that the cement sheath could withstand thermal cycling and mechanical loading over the life of the well.

Post-operation cement evaluation logs confirmed successful placement and zonal isolation with no indications of channeling or fluid migration. These results validated the slurry design and operational strategy, such as the decision to limit string rotation because of high torque to prioritize mechanical integrity over full movement. Finite Element Analysis (FEA) was used to simulate casing-cement-formation interactions and identify potential failure mechanisms under cyclic loading conditions.

A comprehensive overview is presented that discusses the cement strategy applied in the NE field, which integrated laboratory data, field performance, and simulations in specialized software to provide a robust framework for cement operations in complex, ultradeep slimhole environments.

Method, Value Preposition, and Development

Ultradeep Slimhole Definition and Field Description Overview

Oil and gas wells classified as "ultradeep slimhole" are characterized by extreme measured depths that often exceed 20,000 ft (6000 m) and reduced wellbore diameters, typically less than 6 in. in the production section. These configurations are increasingly used to reduce drilling and completion costs, particularly in mature or technically difficult fields. However, the narrow annular clearances in these wells introduce significant complexity during the cement phase, where the achievement of effective zonal isolation becomes a major operational challenge.

The operating company is actively engaged in various operations in the Colombian territory, such as the exploratory project in the country's NE field (Fig. 1). The NE field produces light oil from multiple sandstone reservoirs in the eastern Cordillera, specifically in the Arauca Department. This field produces high-quality light crude oil and natural gas from pay zones composed of a massive sandstone package, with minor interbedded siltstones, shales, and occasional claystones. The grain size ranges from fine to medium, exhibiting consistent sorting. Reservoir intervals show variable permeabilities between 100 and 1000 mDarcy, with porosity values ranging from 9% to 14%.

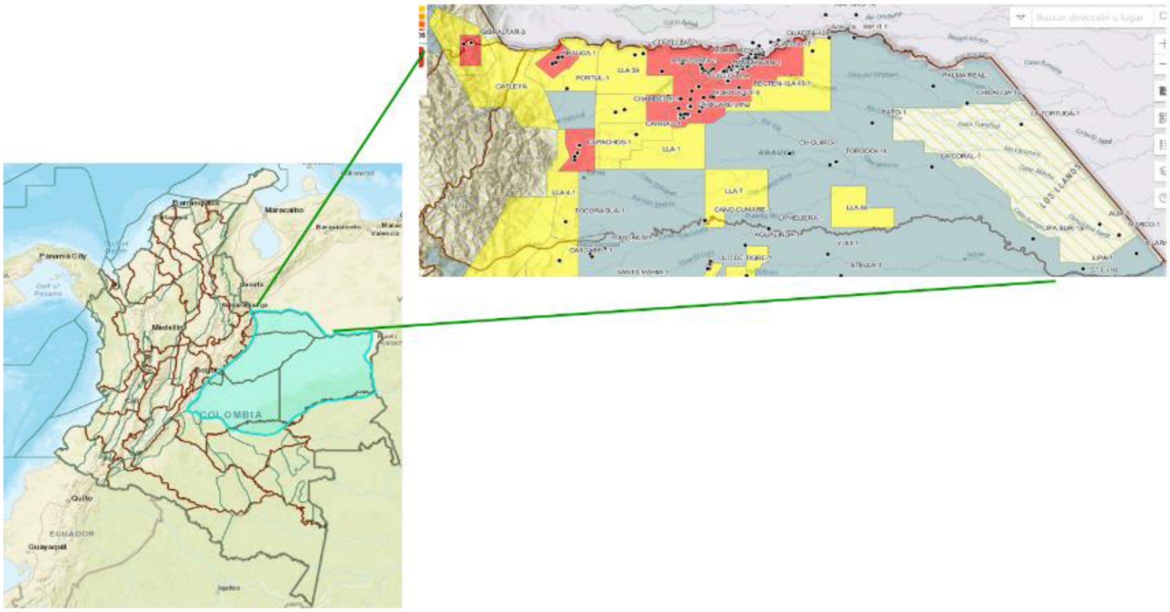


Figure 1—Project location that shows the NE Field.

NE field wells are classified as high-temperature, deep wells (Fig. 2) from regular offset wells drilled in the country often presenting various challenges during cement operations. The well schematic for these wells consists of five-hole sections. Fig. 3 shows the last three sections, where the final section is an ultradeep slimhole.

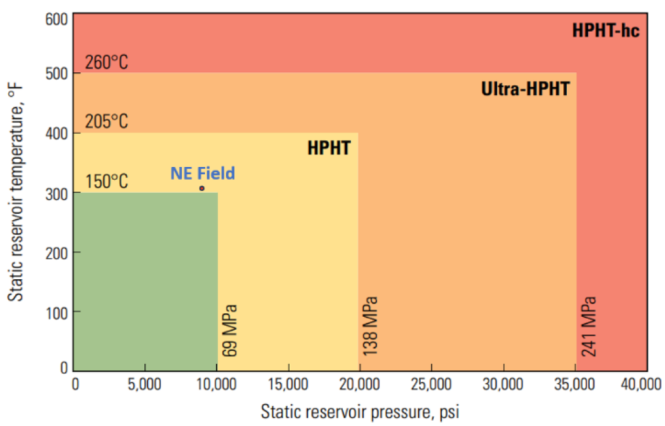


Figure 2—API reference for high-pressure/high-temperature (HP/HT) wells (Smithson 2016).

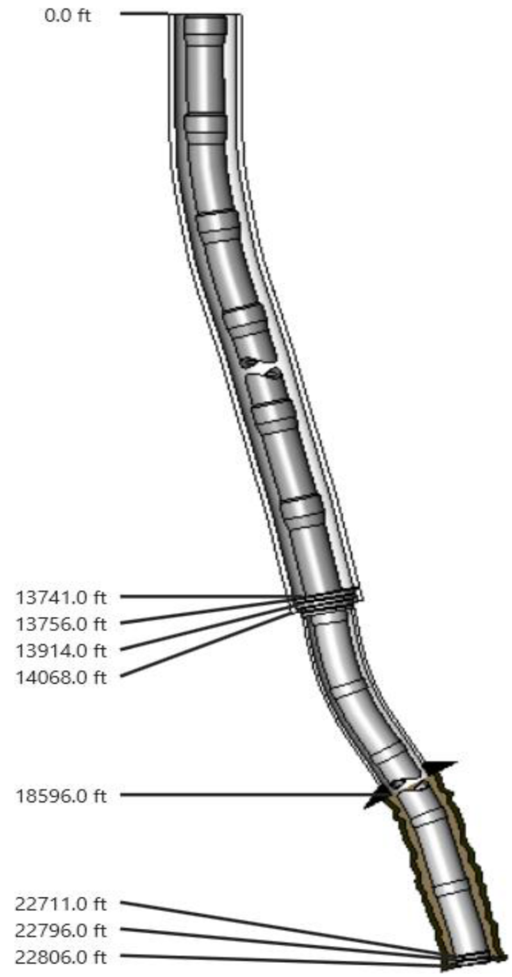


Figure 3—Well schematic for the last three sections.

One of the most critical challenges during cement operations in ultradeep slimhole wells is the high frictional pressure losses that occur during slurry placement. The reduced annular space, combined with long pumping distances, results in elevated ECDs, which can cause formation breakdown or loss of cement integrity. Additionally, high BHST's that often exceed 300°F (150°C) can accelerate slurry thickening and compromise placement time, which requires the use of thermally stable cement systems and advanced retarders. Field studies indicate that even with optimized slurry designs, displacement efficiency is difficult to maintain, and premature setting remains a challenge in these environments (Al-Yami et al. 2009; Salehi and Nygaard 2016).

Geo-mechanical Models

Accurate parameter values along well trajectories in different formations are required to maintain wellbore stability. Key parameters include pore pressure (Po), fracture gradient (Gfrac), collapse pressure (Pc), minimum and maximum horizontal formation stresses, and overburden. Data availability permits the design of the cement operations to achieve planned goals and reduce the potential risk of fluid losses during cement placement and curing. In addition, reservoir petrophysical properties are important to design and execute a cement operation tailored for wells with lost-circulation conditions and narrow operational windows. Table 1 presents a geomechanics model applicable to well construction and cement operations for the phase of interest.

Table 1—Pressure defined in the geomechanical models.

Measured Depth (ft)	True Vertical Depth (ft)	Pore Pressure (psi)	Reservoir Gradient (psi/ft)	Reservoir Density (lbm/gal)	Fracture Pressure (psi)	Fracture Gradient (psi/ft)	Fracture Density (lbm/gal)
	18,551.5	8,002.0	0.431	8.30	14,768.55	0.796	15.32
	18,751.5	8,139.9	0.434	8.36	14,967.70	0.798	15.36
	18,951.5	8,311.4	0.439	8.44	15,185.43	0.801	15.42
	19,151.5	8,377.2	0.437	8.42	15,341.71	0.801	15.42
	19,351.5	8,439.6	0.436	8.40	15,494.89	0.801	15.41
	19,551.5	8,663.9	0.443	8.53	15,736.28	0.805	15.49
	19,751.5	8,602.8	0.436	8.38	15,821.32	0.801	15.42
	19,951.5	8,713.7	0.437	8.41	15,999.15	0.802	15.44
	20,151.5	8,927.7	0.443	8.53	16,239.09	0.806	15.51
	20,351.5	8,743.6	0.430	8.27	16,260.70	0.799	15.38
	20,551.5	9,212.8	0.448	8.63	16,642.57	0.810	15.59
	20,751.5	9,070.7	0.437	8.41	16,689.18	0.804	15.48
	20,951.5	9,087.3	0.434	8.35	16,822.82	0.803	15.46
	21,151.5	9,234.5	0.437	8.40	17,028.46	0.805	15.50
	21,351.5	9,247.5	0.433	8.34	17,158.41	0.804	15.47
	21,551.5	9,372.2	0.435	8.37	17,348.24	0.805	15.50
	21,880.0	9,496.8	0.434	8.36	17,606.96	0.805	15.49

Cement Slurry Design for Ultradeep Slimhole Wells

The design of the cement slurry for ultradeep slimhole wells requires a highly engineered approach to help ensure effective zonal isolation under extreme downhole conditions. The narrow annular clearance significantly increases frictional pressure losses during pumping, which can elevate the ECD and cause formation breakdown. To mitigate this, the slurry should have low plastic viscosity and a controlled yield point that is supported by dispersants and fluid loss additives to maintain flowability and minimize surge pressure.

High BHST that often exceeds 300°F (150°C) requires thermally stable cement systems with advanced retarders to extend thickening time and help prevent premature set. The presence of crossflow between zones further complicates placement and requires sufficient gel strength development and early compressive strength to resist contamination and maintain integrity. Laboratory testing under simulated downhole conditions is essential to validate slurry performance, particularly in rheology, thickening time, and compatibility with drilling fluid (Al-Yami et al. 2009; Salehi and Nygaard 2016).

Premium API Portland cement is essential to meet the technical requirements of the well. Fillers can help improve mechanical properties, reduce Portland cement consumption, and lower the carbon footprint associated with cement manufacturing.

Next-Generation Tailored Spacer System

A tailored spacer is essential for successful cement placement. The synergy between the spacer and cement slurry minimizes the risk of circulation losses and controls fluid loss during pumping. Best cement placement practices require the inclusion of a tailored spacer design in the cement operations plan.

The next-generation tailored spacer system uses additive synergies to help prevent loss circulation in fragile formations and to permit control of rheological hierarchy and wellbore fluid displacement efficiency.

The system can also be tailored with a proprietary blend of lost circulation material (LCM) that can plug fractures up to 3,000 microns wide.

Cement fluid designs should consider rheological and fluid density hierarchies that help improve displacement efficiency and cement bond quality. Design models should also maintain ECD within the operational window defined by pore pressure, collapse pressure, and fracture pressure. Spacer fluid adjustments depend on the expected loss scenario and should follow laboratory-based designs. For seepage or partial-loss scenarios (Arroyave et al. 2023), the spacer is designed to control the loss through the creation of a polymer coating in front of the potential leakage zones caused by high permeability or the presence of microfractures. Fig. 4 illustrates spacer performance under these loss scenarios.

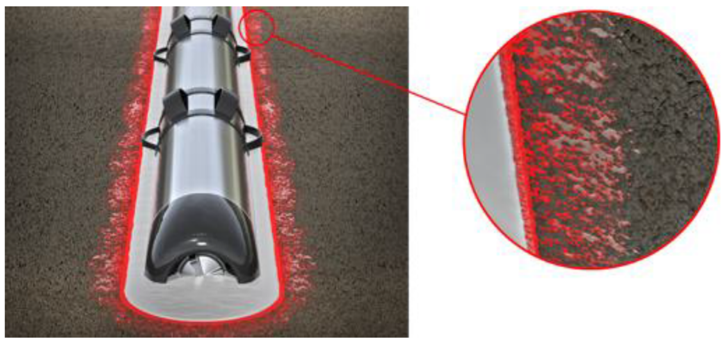


Figure 4—Next-generation tailored spacer.

An analysis of LCM concentration in fluid used for cement operations, such as spacers and slurry systems, is essential to help prevent plugging or bridging the formation in critical flow areas or ports in flotation equipment and configurations with liner overlap. LCM concentration depends on material type, shape, specific gravity, and size distribution, with final values determined through simulations that use specialized algorithms.

Cement Design Engineering

In the Arauca field, this operator's wells are drilled to measured depths greater than 20,000 ft using oil-based mud (OBM) with a density of 9.2 to 9.5 lbm/gal and a 6-in. bit in the production section. This phase is considered a slimhole, with more than 4,200 ft of open hole, followed by a 4.5-in. liner run over a length that exceeds 9,000 ft and anchored into a previously set 9 5/8-in. casing. The geometry results in a reduced annular clearance between the 6-in. open hole and the 4.5-in. liner, as well as an extended annular section of more than 4,800 ft between the inner diameter of a 7-in. drilling liner and the 4.5-in. production liner. By design, the cement top is required to reach the top of the 4.5-in. liner hanger.

Typically, challenges associated with this well design are linked to a high BHST (greater than 300°F) caused by the geothermal gradient, depth, and elevated pressure during cement placement because of frictional losses in the narrow annular spaces. Crossflow can occur because of the close contact between water and oil zones, which can cause channels to form or contaminate the cement slurry in the zones of interest, which potentially results in loss of bond in these areas. The operator applies this mechanical well design strategy with a focus on cost reduction and project feasibility.

The cement operation uses a 16.0-lbm/gal tailored cement slurry designed for high BHST conditions. The base slurry formulation should meet specific performance criteria to help ensure effective zonal isolation within the annulus and maintain long-term integrity throughout the life of the well. Laboratory tests under simulated downhole conditions validate the slurry's behavior, particularly in terms of compressive strength development and avoidance of the retrogression phenomena observed in Portland cement systems. Rheological control is critical to minimize frictional pressure losses that could elevate the ECD beyond the formation fracture gradient. A short transition time is achieved by using a preset expansion agent, while the slurry should resist to settling under downhole conditions and remain compatible with the drilling fluid system.

Additionally, the evaluation of mechanical properties, such as YM and PR, is used to assess the system's resilience and capability to provide long-term integrity. The cement system is also designed to minimize the carbon footprint through a reduced use of premium Portland cement in the formulation.

During these operational designs, fracture pressure, minimum stress, pore pressure, and collapse pressure values from the geo-mechanical model for the Arauca field were applied during the operational design. These parameters were entered into advanced cement simulation software to calculate wellhead pressure, pump rate, cement fluid volume, and ECD, as well as to identify integrity risks near formations susceptible to circulation losses. Fig. 5 shows the fluid position inside a well in the NE field.

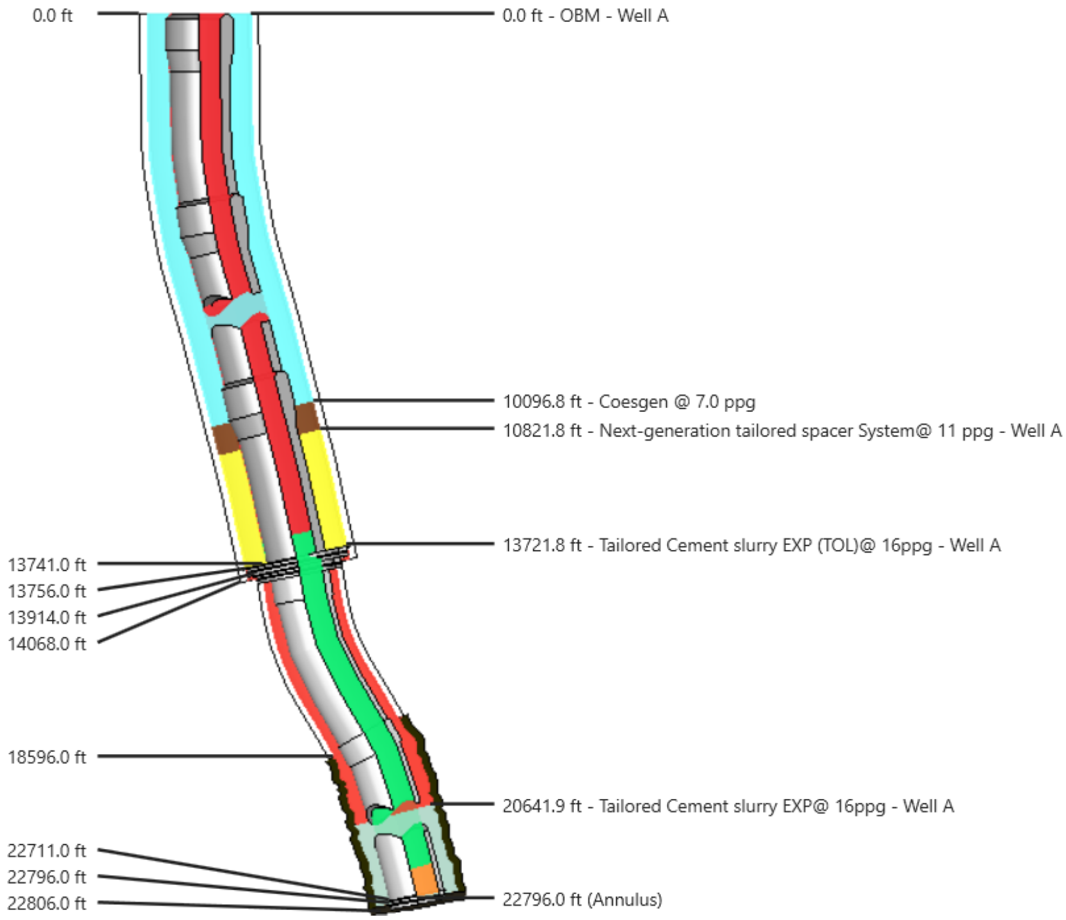


Figure 5—Planned cement fluid position.

Laboratory Tests. Tests, such as thickening time, free water, rheology, BP settling, static gel strength (SGS), compressive strength, tensile strength, and mixability are important to any cement design. However, the tests with the greatest influence on the design of ultradeep, high-temperature slimhole wells under potential inflow conditions are examined.

Rheology Guidelines for Cement Design. In a slimhole environment, the necessity for a low-rheology cement slurry at well conditions is evident. Excessive slurry viscosity can produce high pressure during placement, which may result in formation breakdown and compromise zonal isolation. The primary design factor related to viscosity is to achieve the lowest possible rheology without the solids settling. The addition of too much dispersant can be as serious as insufficient dispersant. An over-dispersed slurry can release an excessive breakout of supernatant water from the slurry, which, in the small annular volumes common in slimhole completions, can translate into a large linear annular distance.

Compressive Strength (UCA). Compressive strength is a mechanical property that measures the ability of a material, such as set cement, to withstand axial loads without failure. In oil and gas well cement operations, this parameter is critical for the evaluation of the structural integrity of the cement sheath after curing. It is typically expressed in pounds per square inch (psi) or megapascals (MPa) and is determined through laboratory tests under simulated downhole conditions.

In high-temperature wells, extreme thermal conditions affect the compressive strength of cement because they alter the hydration process and the microstructural development of the cement matrix. At temperature greater than 230°F, Portland cement undergoes mineralogical transformations in the compounds that control early and late compressive strength, namely tricalcium silicate (C_3S) and dicalcium silicate (C_2S), respectively. These changes cause a loss of integrity in the cement sheath, a phenomenon known as "retrogression."

To maintain adequate strength under high-temperature conditions, specially formulated cement systems are required, typically with a calcium-to-silica ratio (C/S) near unity. This balance preserves long-term compressive strength and helps ensure reliable zonal isolation.

Mechanical Properties Analyzers (MPRO). In the context of cement operations, MPRO testing refers to a series of laboratory evaluations that assess the mechanical integrity of cement slurries after they set. These tests measure key properties, such as:

- Compressive strength: capability of the cement to withstand axial loads
- Tensile strength: resistance to tensile forces, often evaluated indirectly
- YM: measurement of stiffness that indicates the extent of deformation under stress
- PR: ratio of lateral strain to axial strain under load

These mechanical properties are essential to help ensure effective zonal isolation, casing support, and long-term well integrity, particularly in HP/HT environments. MPRO tests are typically conducted using specialized mechanical test equipment after the cement has cured under simulated downhole conditions.

Static Gel Strength. In a true fluid system, hydrostatic pressure remains present and stable. The potential loss of hydrostatic pressure transmission is directly proportional to the SGS level, which is a dynamic and progressive physical property. The cement column's length and diameter also influence the extent of hydrostatic pressure loss.

This property is primarily used to evaluate the transition time (ideally less than 45 minutes), during which the cement slurry evolves from a Critical Static Gel Strength (CSGS) usually 100 lbf/100 ft^2 to 500 lbf/100 ft^2. This transition is crucial to assess gel strength development, which directly impacts fluid migration control and wellbore stability during the early stages of cement setting.

Fig. 6 illustrates the calculation of CSGS measured to control fluid migration caused by water-oil contact in an ultradeep slimhole well.

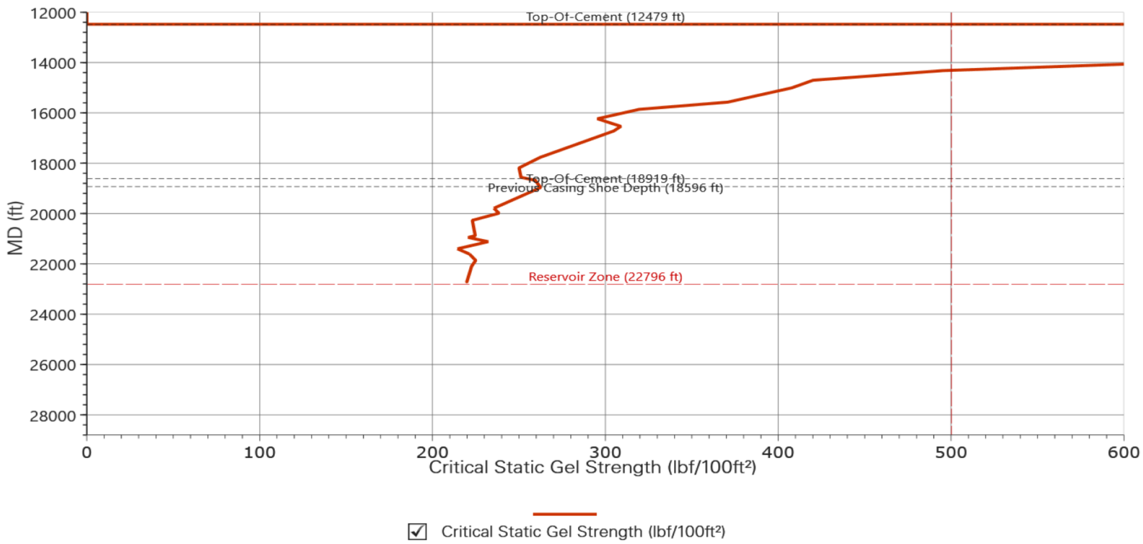


Figure 6—Critical static gel strength.

Table 2 presents a summary of the properties and laboratory test results.

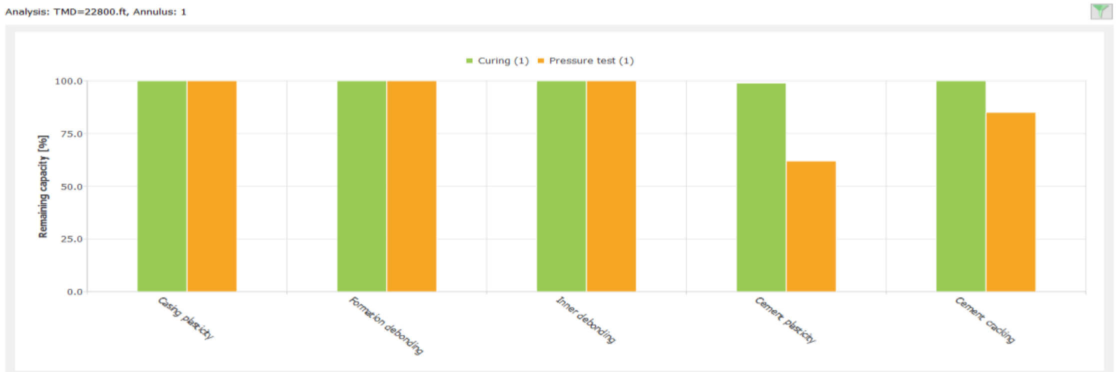
Table 2—Tailored slurry properties for an ultradeep slimhole well.

Requirements	Results
Energized slurry/density (lbm/gal)	16.0
API rheology(190°F, τ lbf/100 ft^2)	Θ:300, τ:167, Θ:200, τ:123, Θ:100, τ:67, Θ:60, τ:42, Θ:30, τ:22 Θ:6, τ:6 Θ:3, τ:4
Fluid loss (cc/30 min)	30 (measured on the base slurry)
Compressive strength (UCA) 24 hr, psi (315°F)	2,610
Critical static gel strength	15 min – 200 lbf/100 ft^2 to 500 lbf/100 ft^2
Free water (%)	0
Sedimentation API (lbm/gal)	< 0.5
Thickening time (hrs: min)	05:30
YM (Mpsi)/PR	1.3 /0.21

Long-Term Analysis. When the narrow annular space of slimhole completions is cemented, a correspondingly thin cement sheath results. Intuitively, as the cross-sectional area of the sheath decreases, its overall mechanical integrity can also be reduced. Most available literature addresses the compressive strength required for zonal isolation but does not differentiate between thin and conventional sheath geometries.

Additional mechanical properties, such as YM and PR, should factor into the design a more ductile and less brittle cement system. An optimal compressive strength of approximately 2,000 psi, a YM near 1 million psi, and a PR greater than 0.2 could be relevant values for analysis of thermal cycling and load cycles throughout the life of the well.

The cement's static mechanical properties will be measured using triaxial cells. The FEA will then evaluate failure criteria in the casing-cement-formation interaction during well life cycles. Fig. 7 presents the FEA results for the cement slurry system designed and implemented in the Arauca wells.



	Curing (1)	Pressure test (1)
Casing plasticity	100	100
Formation debonding	100	100
Inner debonding	100	100
Cement plasticity	98.9	62
Cement cracking	100	85

Figure 7—FEA results.

Basically, it maintains a certain amount of linear elastic capacity available for use before the onset of nonlinear material behavior. This condition was observed after the simulation of cyclic pressure loads during a 10-year period.

Results, Discussion, and Conclusions

Cement Operation Execution: Field Application

For practical purposes, Well A serves as the reference case. The production liner section was drilled with a 6-in. PDC bit to a final depth of 22,806 ft MD (21,883 ft TVD) using an OBM with final fluid properties of 9.2 lbm/gal, yield point (YP) of 11 lbf/100 ft^2, and a plastic viscosity (PV) of 20 cP. A 4.5-in. production liner was then run to 22,796 ft MD and cemented using a tailored cement slurry formulated with a preset expansion agent for ultradeep slimhole applications. The slurry had a density of 16.0 lbm/gal and was placed from the bottom hole to the top of the liner at 13,741 ft.

The 16.0-lbm/gal tailored slurry properties were evaluated via laboratory tests (Table 2). The thickening time was adjusted with an operational safety factor of 1 hour and 50 minutes to allow sufficient placement time. The slurry design maintained a clear rheological hierarchy for both the drilling mud and spacer, which allowed efficient displacement. The compressive strength exceeded 2,000 psi at bottomhole conditions, which confirmed the system's mechanical integrity under ultradeep slimhole environments.

The slurry design considered the gas migration theory.Development of SGS (i.e., internal rigidity within the cement matrix) usually begins shortly after pumping ceases and continues until the cement sets. As gel strength develops, the cement column partially supports itself. The hydrostatic pressure of the column decreases as its self-support increases and is directly proportional to any volume changes that occur when filtrate is lost from the matrix of the cement slurry into the formation or when minor hydration-related volume reduction occurs as the cement sets.

Formation fluid migration (gas or liquid) can result from hydrostatic pressure loss. The capability to achieve sufficient gel strength in the early setting process can help resist fluid flow. Per API RP65 Part 1 or RP 65 Part 2, the CSGS value is defined as the point where hydrostatic pressure decreases until the wellbore reaches equilibrium with pore pressure at the depth of the potential flow zone, which then allows the influx of gas or liquid.

The fluid loss and transition time measurements of a cement systems help determine the cement systems ability to prevent fluid or gas flow during and immediately following placement of the cement system.

A cement system with fluid loss of less than 50 cc/30 min and less than a 30-minute transition time is recommended to prevent flow through unset cement (B. E. Clark et al 2023). The cement system designed builds gel strength rapidly and provides a fast transition time (100-500 lb/100 sq ft) which limits the amount of time the cement slurry is susceptible to flow through the unset cement. The system can deliver transition times of less than 30 minute.

Furthermore, a next-generation tailored spacer with a density of 11 lbm/gal was used. The formulation helped ensure compatibility with the OBM and altered the wettability of the formation face to water-wet conditions to help improve post-set cement bond quality. The next-generation tailored spacer design included LCM to help prevent any lost circulation.

This section of the well, presented several operational challenges, such as an undisturbed BHST of 315°F (Fig. 8), an elevated risk of lost circulation during cement placement, and the requirement to achieve effective drilling fluid displacement within a narrow ECD window relative to the fracture pressure, particularly under conditions of poor casing centralization.

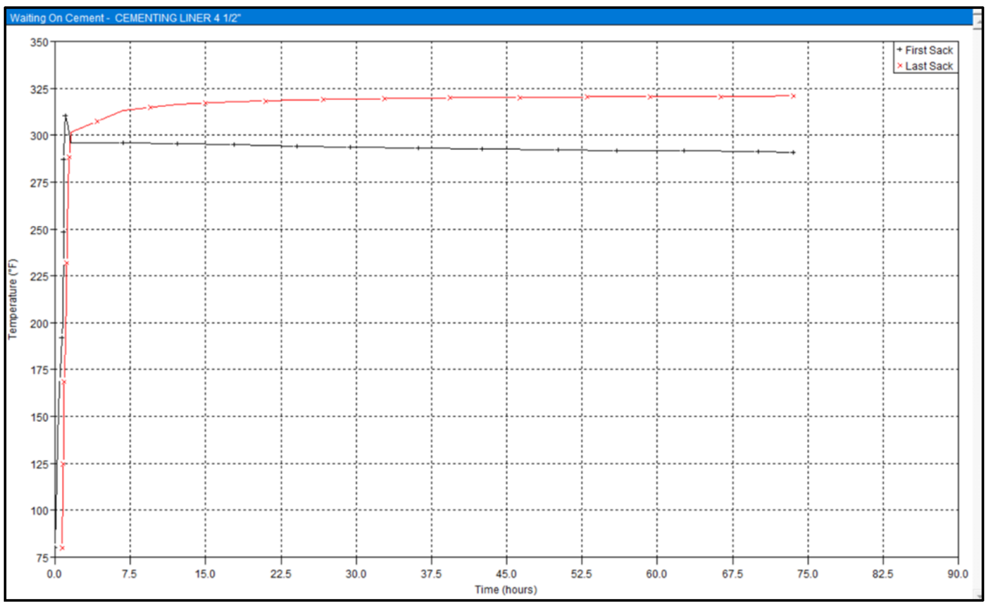


Figure 8—Temperature simulated in specialized software.

Additional objectives included prevention of channeling from influx, mitigation of post-placement fluid migration, reduction of annular pressure buildup throughout the life of the well, placement of top of cement at the liner hanger setting depth, and formation of a competent cement sheath confirmed through CBL evaluation. Pipe movement improved displacement efficiency during cement operation execution.

Well A followed the cement program. Cement fluids were successfully pumped without fluid losses, and the system registered an end circulation pressure of 4,000 psi before the plugs bumped. The plugs that contacted the landing collar confirmed proper fluid displacement into the annulus. After the tool released from the liner hanger and circulation resumed at the liner top, visual inspection of returns confirmed the presence of excess spacer. Fig. 9 summarizes key parameters from the cement operations in Well A, such as pump pressure trends, fluid densities, pump rates, and fluid volumes.

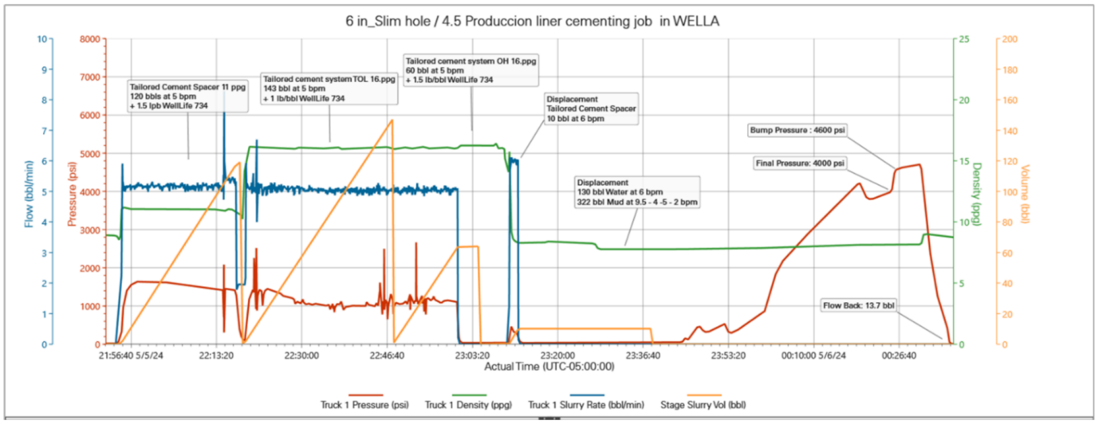


Figure 9—Key parameters from the executed cement operations in Well A.

Presentation of Data and Results

In Well A, cement fluids were pumped correctly based on the cement program with no observed losses. The dart and wiper plug latch was identified at the expected depth and volume. Plug contact with the landing collar confirmed completion of fluid pumping and placement in the annulus. After the tool released from the liner hanger and circulation resumed at the top of the liner, returns showed excess spacer.

The ECD parameter was controlled in real time during fluid cement pumping to help prevent exceeding the fracture gradient. The operation recorded a maximum ECD of 12.98 lbm/gal, which closely matched the value simulated in the cement operation plan. No losses occurred during the cement operation, which indicated the reliability of the slurry system and the next-generation tailored spacer system to control these conditions during fluid placement. Fig. 10 shows the ECD parameter for the cement operation.

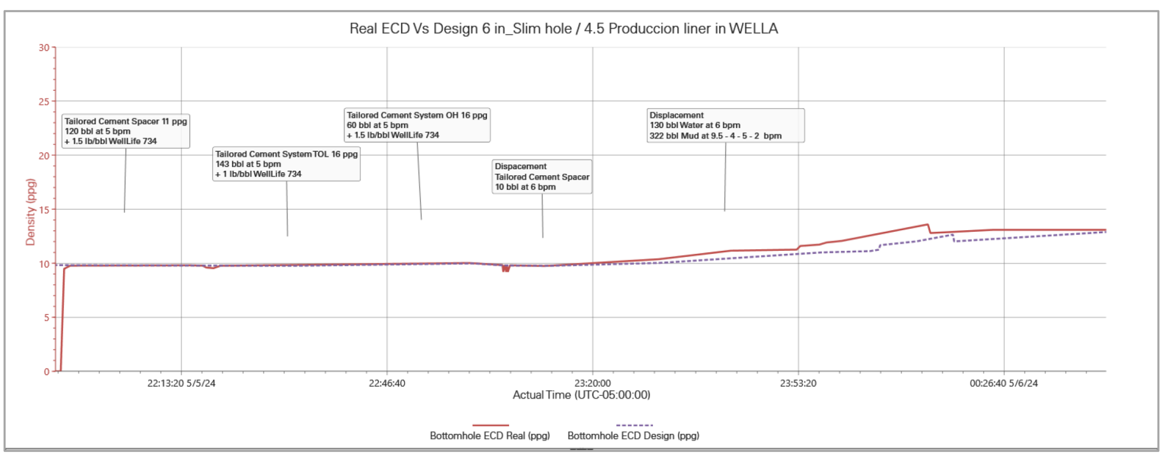


Figure 10—Real-time ECD measured during the cement operation.

During primary cement operations, operators typically pump at high rates to induce turbulent flow to facilitate mud displacement and filter-cake removal. However, rates should remain within limits to help prevent excessive bottomhole ECDs and pressure that could cause lost circulation, exceed the formation fracture pressure, or create excessive surface pressure.

Even in conventional operations, this balance is complicated by significant density differences between the mud being displaced, the cement, and the displacement fluid. These differences can produce "free fall" conditions that make accurate prediction of true bottomhole pressure more difficult. In slimhole cement operations, the problem compounds because smaller tubulars and reduced annular clearances generate higher friction pressure and related ECDs for the same pump rate.

Pipe movement (rotation and reciprocation) remains a highly recommended practice to help improve mud displacement. However, tight clearances in slimhole configurations amplify surge-and-swab effects. Downward casing movement during reciprocation exerts a piston-like force on the fluid, which increases bottomhole pressure (surge). This condition increases sensitivity to excess formation fracture pressure and triggers lost circulation. Upward pipe movement during reciprocation can reduce bottomhole pressure, potentially to the extent that allows formation fluid to enter the wellbore. In either case, the effectiveness of cement operation execution and performance (cement bond) can be compromised. These effects require explicit evaluation during slimhole cement procedure development.

During the cement operation, the string could only rotate partially because of high torque observed during the operation. The operation prioritized preservation of the string components and connection integrity until successful completion.

To corroborate the operational results, Fig. 11 presents the log evaluation from the previous operation. The log corresponds to the sonic wave measurements, the CBL, and the variable density log. The results show complete zonal isolation in the openhole section and excellent adherence to both the casing and formation.

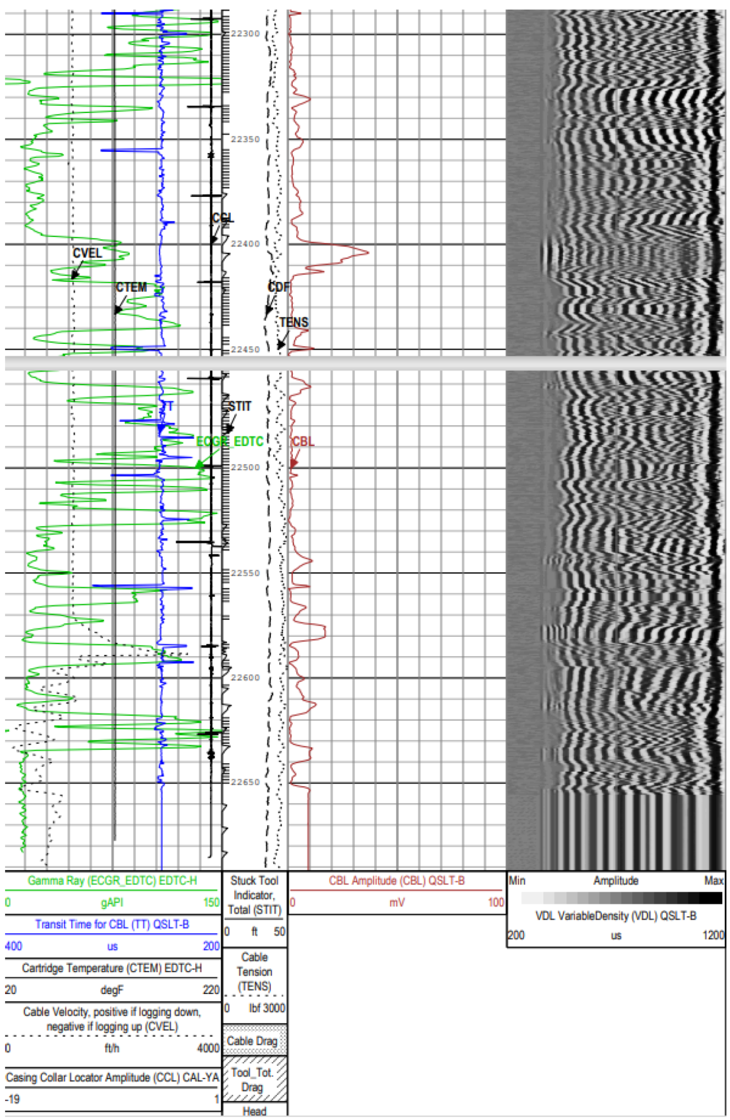


Figure 11—Cement evaluation log for the 12,300- to 22,700-ft interval.

Conclusions

The cement strategy applied in the Arauca field demonstrates the effectiveness of customized slurry systems to address the unique challenges of ultradeep slimhole wells. Optimization of rheological behavior and mechanical properties allowed reliable placement, reduction of frictional losses, and preservation of integrity under thermal and mechanical stress.

Expansion agents and short transition time formulations improved sealing performance in zones affected by crossflow. Cement evaluation logs confirmed the absence of debonding across the pay zones.

Laboratory tests and the FEA provided critical insights into long-term behavior, which included resilience under cyclic loading and thermal cycling. Operational decisions, such as the prioritization of string integrity over full rotation, contributed to the operation's success.

Overall, the importance of the integration of material science, simulation, and field experience to improve cement operation performance in complex well environment was emphasized in this study.

Nomenclature

API	American Petroleum Institute
bbl	barrels
bbl/min	barrels per minute
BHST	bottomhole static temperatures
CBL	cement bond log
ECD	equivalent circulating density
ft	feet
Gfrac	fracture gradient
lbm	pounds
lbm/gal	pounds per gallon
MD	measured depth
Po	pore pressure
PR	Poisson's ratio
psi	pounds per square inch
TVD	true vertical depth
YM	Young's modulus
Θ	shear rate
τ	shear stress

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